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A low-power Wave Union TDC implemented in FPGA

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ABSTRACT: A low-power time-to-digital convertor (TDC) for an application inside a vacuum has been implemented based on the Wave Union TDC scheme in a low-cost field-programmable gate array (FPGA) device. Bench top tests have shown that a time measurement resolution better than 30 ps (standard deviation of time differences between two channels) is achieved. Special firmware design practices are taken to reduce power consumption. The measurements indicate that with 32 channels fitting in the FPGA device, the power consumption on the FPGA core voltage is approximately 9.3 mW/channel and the total power consumption including both core and I/O banks is less than 27 mW/channel.

KEYWORDS: Front-end electronics for detector readout; Digital electronic circuits

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1 Introduction

Chain structure in existing FPGA families can be used for time-to-digital conversion (TDC) purposes [1]–[13]. The Mu2e experiment [14] at Fermilab requires operating a large channel-count straw tube chamber system inside vacuum demanding low-power consumption on all stages of readout electronics including the TDC device. The time difference of signals detected at both ends of a straw tube is measured to a precision better than 35 ps (rms) so that the location of the charged particle hit along the tube length can be determined. This required time resolution is much finer than that of typical straw tube TDC in which 1–2 ns is sufficient.

The Wave Union TDC [7] is a scheme developed in our previous work to improve the resolution of the TDC implemented in the FPGA beyond its cell delay. Multiple 0–1 and 1–0 transitions are generated in the delay chain in the Wave Union TDC and registered for encoding, which effectively provides multiple measurements with one set of delay chain and register array structure and thus improves time measurement resolution. (Note that regular TDCs make one measurement with a single 0–1 transition.) Intrinsically, the Wave Union TDC is a low-resource and low-power consumption scheme since in order to achieve a finer resolution, less logic elements are used in the Wave Union TDC comparing to regular TDC schemes making one measurement with a single 0–1 transition.

In addition, special digital design practices are taken to properly interface the fast clock (250 MHz) and the slow clock (62.5 MHz) domains so that in each channel, the fast clock is confined in the register array used to capture wave union transitions in the delay line, while most

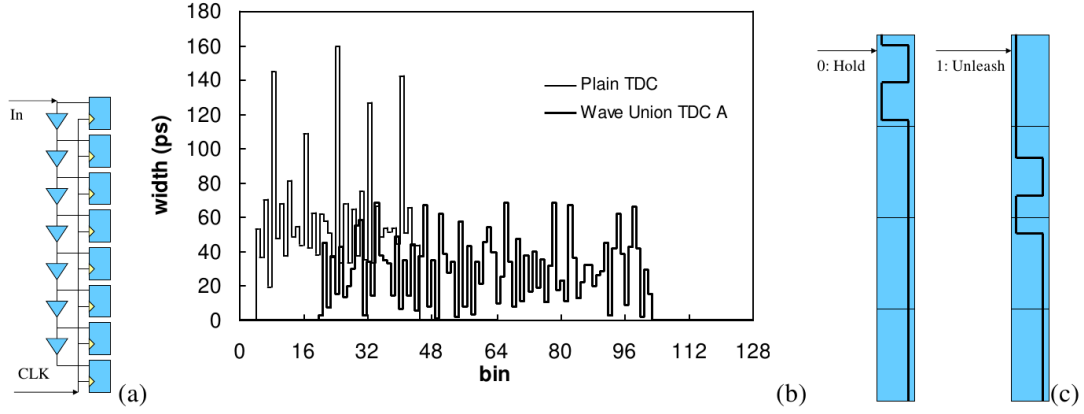


Figure 1. A delay chain based TDC structure (a), the bin width plot (b) and a wave union TDC structure (c).

encoding and data handling blocks are put in the slow clock domain. Taking advantage that the required minimum double hits separation is relatively large (no less than 64 ns), encoders and other common functional blocks are shared among multiple channels to further reduce logic element usage and power consumption.

In this paper, the Wave Union TDC is first discussed in section 2 followed with descriptions of the digital design practices utilized in the low-power edition of the 32 channels TDC in section 3. Test results of the standard deviation of the time measurements and the power consumption are presented in section 4. Future work and further improvements are discussed in section 5.

2 The Wave Union TDC

2.1 Imperfectness of delay chains inside FPGA

A delay chain based TDC consists of a delay chain followed by a register array as shown in figure 1(a). A special issue of the delay chain based TDC implemented in FPGA is its large differential nonlinearity (DNL) as shown in the histogram marked with “Plain TDC” in figure 1(b) which is represented as apparent width of each TDC bin.

The most significant origin of DNL is the logic array block (LAB) structure. When the input signal in the carry chain passes across the LAB boundaries (and also the half-LAB boundaries in some FPGA families), extra delays added cause periodic “ultra-wide bins.”

2.2 The Wave Union TDC scheme

In our previous work [7], an approach called the “wave union TDC” was developed to sub-divide the ultra-wide bins and to improve measurement resolution. The key part in the wave union TDC is the “wave union launcher” as shown in figure 1(c). A set of specially designed logic described in [8] “holds” a “wave union” with several 0-to-1 or 1-to-0 logic transitions in the delay chain when the input level is 0 and “unleashes” the wave union when the input level becomes 1. It feeds the wave union into the TDC delay chain/register structure, making multiple measurements. We used two measurements from two 1-to-0 transitions and the TDC codes (i.e., the registered positions) of the two transitions are summed into a new bin number. As shown in figure 1(b), effectively

the “ultra-wide bins” are subdivided using the wave union TDC scheme. Since the bin number for the Wave Union TDC is the sum of the TDC codes of two single transitions, it shows a wider range in figure 1(b). In actual design, the delay chain used here is the carry chain existing in the FPGA. Therefore manual placements are not necessary for building the delay structure but can be optionally applied for more evenly spread of the logic element usage inside FPGA.

3 Digital design practices for low-power consumption

For operating in a vacuum, several digital design practices are taken so that the TDC consumes less power without degrading important performance requirements. The Wave Union TDC scheme itself is a power saving scheme. Additionally, in the FPGA environment, higher clock frequency causes more power consumption in the internal clock distribution network. We have tried using a high frequency clock only in the necessary portion of firmware while keeping most other portions with a clock at lower frequency. Data flow sequence is carefully designed so that some functional blocks can be shared among multiple channels, which reduces silicon area usage and therefore power consumptions. These practices will be discussed in the following.

3.1 Wave Union TDC as a power saving scheme

The TDC resolution can be improved by ganging several channels together, i.e., sending an input into multiple channels. When ganging several TDC channels together, multiple measurements provide more information of the input time.

In the wave union TDC scheme, multiple measurements are made in a single set of the delay chain, register array, encoder and other logic functional blocks. Logic element usage and therefore, power consumption is clearly smaller than ganging channels together. For a similar time resolution without wave union, more logic elements are needed.

3.2 Clock domain arrangement

The sampling registers are to be operated in a relatively high frequency so that the length of the delay chain is not too long. In our TDC firmware, the sampling registers are driven by a 250 MHz clock as shown in figure 2.

On the other hand, encoders and other stages of the data processing do not need to operate at that high frequency given that the hit rates in the TDC channels are not too high. In our firmware, encoders and other stages are clocked at $250/4 = 62.5$ MHz to reduce power consumption. To interface the two clock domains, a clock disable sequencer disables the sampling registers for 15 clock cycles (at 250 MHz) after registering a valid hit, so that the outputs of the sampling registers become stable for exactly 64 ns. The data load/transfer registers are enabled to load data every 64 ns (4 clock cycles at 62.5 MHz) so that any valid hit can be loaded once. After loading, the data are transferred downward one by one.

This scheme restricts the separation of double hits to be at least 64 ns, and this is acceptable in our application.

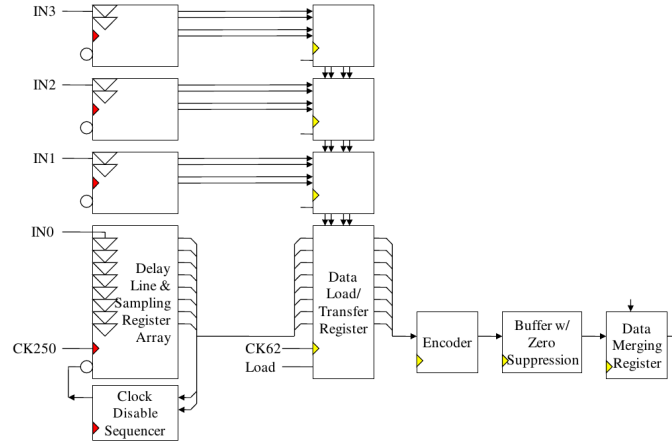


Figure 2. Block diagram of a four-channel group.

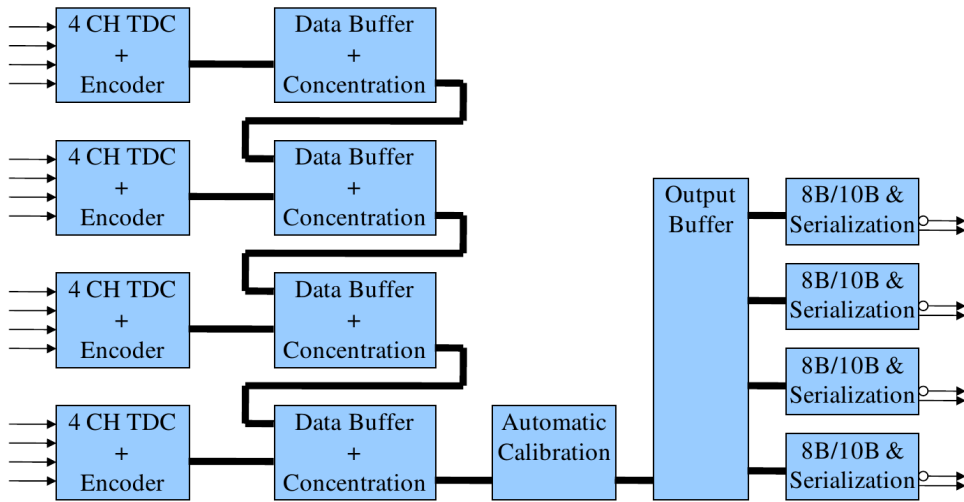


Figure 3. Block diagram of a partial (16 channels) TDC firmware.

3.3 Silicon resource sharing

Note that if the data load/transfer registers were enabled to load every clock cycle, the double hit separation would be improved to 16 ns. The reason of choosing 64 ns is to permit further resource saving as discussed in this section. In every four clock cycles, the data load/transfer registers are enabled to load one clock cycle for the valid hit data while transferring data from other channels in the remaining three cycles. This will allow four channels to share one encoder and one zero suppression buffer as shown in figure 2. In the larger scale, the similar structure is also utilized as shown in figure 3.

In figure 3, four sets of four-channel groups, 16 channels total, are connected in a daisy chain style. In each channel group, zero suppressed data are stored in the data buffer. Data are readout from the buffer and fed through the automatic calibration block. The automatic calibration block accumulates a DNL (or bin width) histogram of the raw TDC data for each channel, converts the DNL histogram into a lookup table and the lookup table is used to translate the TDC code into the

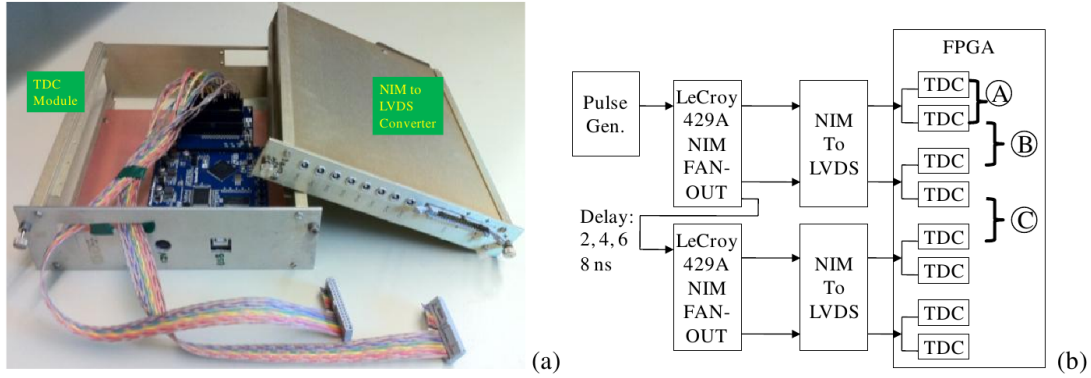


Figure 4. Photograph of the test module (a) and test environment (b).

hit time with bin width calibrated. The detailed operation sequence of the automatic calibration block can be found in [7]–[9] and [11]. The calibrated data are sent into the output buffer. The data in the output buffer are distributed to four 8B/10B serialization blocks and sent out of the FPGA via 4 differential pairs at 250 M bits/s each.

With careful resource saving design practices in various places, 32 TDC channels are fit into the FPGA, taking 56% out of 24624 logic elements in the EP3C25F324C6N device.

4 Test results

The low-power edition of the Wave Union TDC has been developed in a low-cost Altera Cyclone III FPGA device and tested in an evaluation board as shown in figure 4(a).

A NIM pulse is generated and fan-out units are used to create 8 copies of the pulse, 4 of which come from the first fan-out unit and another 4 from the second fan-out unit with the input delayed from the first unit by 2, 4, 6 and 8 ns. The NIM pulses are converted into LVDS signals and interfaced with the FPGA. Inside the FPGA, 16 channels TDCs are ganged into 8 groups, 2 channels each. There are three types of time differences as marked A, B and C in figure 4(b). They are time differences between (A) two ganged channels, (B) two channels from the same NIM fan-out unit and (C) different NIM fan-out units. The test results of these time differences are described in the following.

4.1 Time measurement resolution

Typical histograms of three types (A, B and C) of time differences described above are shown in figure 5(a) (marked with HistA, HistB and HistC). Arbitrary offsets are added to these histograms so that they can be plotted in a single picture. The standard deviations of time differences between various channel pairs are plotted in figure 5(b) with each data point representing a channel pair.

When the two channels are ganged together inside the FPGA (type A), the time differences between them are less than 100 ps reflecting primarily the skews of different channels and are independent of the external cable delays. The standard deviations are mostly in the range of 20 to 30 ps as shown in the left portion ($dt < 100$ ps) of figure 5(b). When the time differences are measured from two channels connected to the same fan-out unit (type B) the values of time differences are

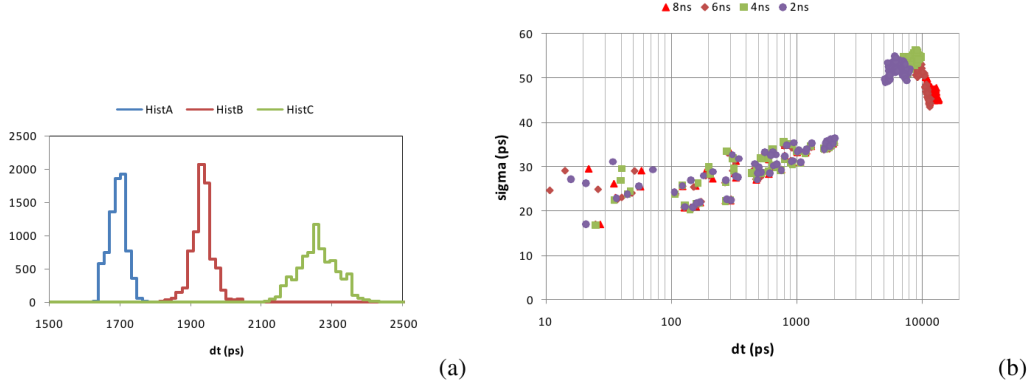


Figure 5. Typical time difference histograms (a) and standard deviations vs. time differences (b).

from 100 ps to 2 ns primarily due to the skews of different channels and are independent of the external cable delays. It is the closest configuration to the regular in-system applications, and the standard deviations are from 20 to 35 ps as shown in the middle portion ($100 \text{ ps} < dt < 2 \text{ ns}$) in figure 5(b). The type C time differences represent the cable delay (2 to 8 ns) plus a delay of the fan-out unit so that their values of dt spread approximately from 5 to 13 ns. The standard deviations actually show the additional jitters of the additional NIM fan-out unit and have total values around 50 ps as shown in the right portion ($dt > 4 \text{ ns}$) in figure 5(b).

It can be seen that even under conditions of using switching power supplies provided on the evaluation board and having single ended signal connections between the FPGA and other devices, a time measurement resolution of 30 ps is achieved.

4.2 Effects of power supply noise

It is known that the power supply noise degrades TDC time measurement resolution severely. The power voltages needed for the FPGA, i.e., 1.2 V for the core and 2.5 V for the I/O banks are generated with a switching power converter on the evaluation board and the tests described above are performed under this condition. The power nets on the evaluation board can be disconnected and cleaner linear power supplies can be connected to achieve better time measurement resolutions. A comparison of the different power supplies is made.

The standard deviations of the time differences between two channels are measured for different relative timings. The standard deviations are approximately 30 ps when using a switching power supply and are approximately 25 ps when using a cleaner linear power supply. As expected, time measurement resolution improves when the power noise is reduced.

4.3 Power consumption

To evaluate power consumption, 32 TDC channels are fit into the FPGA and currents in power nets for both FPGA core and I/O bank are measured.

In normal operating conditions, the current in the 1.2 V net supplying the FPGA core is 0.23 to 0.25 A that yields 276 to 300 mW for 32 channels i.e., 8.6 to 9.3 mW/channel. The 2.5 V I/O bank power supply draws 0.17 to 0.22 A that consumes 425 to 550 mW for the entire chip. Therefore in

this test, the total power consumption is around 22 to 27 mW/channel, including both FPGA core and I/O banks.

5 Discussions

For most applications in low-radiation environments requiring time measurement resolutions of 30 to 100 ps in nuclear physics, high energy physics and medical physics, FPGA TDCs become feasible due to their low-cost and flexibility. The power consumption can be reduced with various design practices. It can be seen that in our firmware, the core power consumption has already been reduced significantly lower than the one in the I/O banks with 2.5 V I/O standards (LVDS, 2.5VCOMS etc.). Further reduction of total power can be anticipated when I/O standards at lower voltage are utilized.

Acknowledgments

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